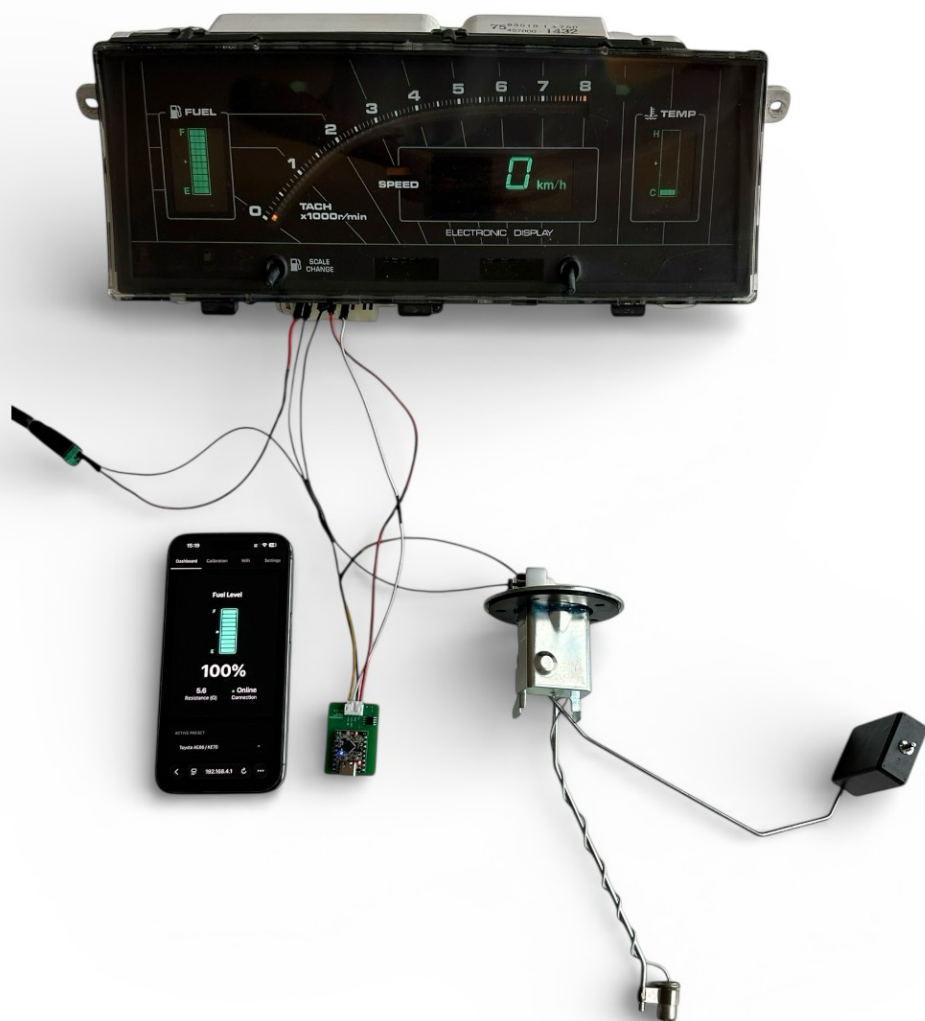


DIGITAL FUEL LEVEL METER CONTROLLER



Version 1.1

FASTROAD

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IMPORTANT DISCLAIMER I am **not a professional electrical engineer**, I'm an automotive engineer who enjoys tinkering with electrical parts as a hobby.

This module is for people who want to upgrade their classic car to have a digital dash and would like to keep the stock fuel level sender, or for people whose original digital fuel level sender is broken, unavailable, or too expensive to replace.

Use at your own risk. Always double-check connections, verify voltages, and test in a safe environment. If you're unsure about any electrical work, consult a qualified auto electrician.

If you plan to reprogram/customize the output range yourself (via the web application interface), **take the fuel level sender out of the tank** to make the job easier and more accurate. This lets you manually move the float arm to the exact 0%, 50%, and 100% positions. Tank levels can vary slightly due to age, sender wear, rust, so don't rely solely on published values. Making a custom range gives the best results for your specific car.

This module was developed and tested primarily with a **Toyota AE86/KE70** fuel level meter in a digital dash setup, but the stock sender resistance range is quite similar across any 80's Toyotas.

1. Introduction & Product Overview

Thank you for purchasing the Programmable Fuel Level Converter Module!

Why this exists: Original JDM digital fuel level meters (and compatible senders for the rare AE86 digital instrument clusters) are no longer in production by Toyota. The few remaining NOS or aftermarket replacements are extremely hard to find and very expensive. This makes full digital dash upgrades prohibitively costly for many enthusiasts.

This compact, affordable converter solves that problem by letting you keep your stock fuel sender while making it work perfectly with modern/aftermarket digital dashboards that expect a different input signal.

The stock Toyota AE86/KE70 fuel level sender (in-tank float unit) uses a resistance range of approximately:

- **Full tank** → $\sim 3\ \Omega$ (ohms)
- **Empty tank** → $\sim 110\ \Omega$

Many aftermarket digital dashes expect a voltage signal of 0-5V, so this module acts as an intelligent bridge: it reads the original sender resistance and outputs a programmable/converted signal that your digital dash can understand correctly.

It comes pre-loaded with a preset map optimized for Toyota AE86/KE70 compatibility, but you can change it upon request or program it yourself via the web application.

2. Features

- Fully programmable output range via web application
- Compatible with any digital cluster that uses 0-5V input
- Tiny form factor – fits easily behind the dash
- Simple 4-wire connection
- Upload/Export custom presets

3. What's Included

- 1 × Programmable Fuel Level Converter PCB
- Pre-soldered connector with wires
- Protection box
- This manual (digital/PDF version recommended)

4. Installation & Wiring

Safety first: Disconnect the battery before working on any electrical components.

This module was designed to be used behind the cluster, since the factory wires make hookup much easier that way.

1. Connect the module as shown below:
 - **5V** → to a switched 5V source (from digital cluster, check image bellow)
 - **GND from controller** → to fuel level meter ground (if unsure check fuel level pins for continuity with car ground)
 - **GND from plug** → to any place, near the cluster (chassis ground)
 - **FUEL IN** → to the wire coming from the stock sender (signal wire)
 - **FUEL OUT** → to the fuel input pin on your digital dash

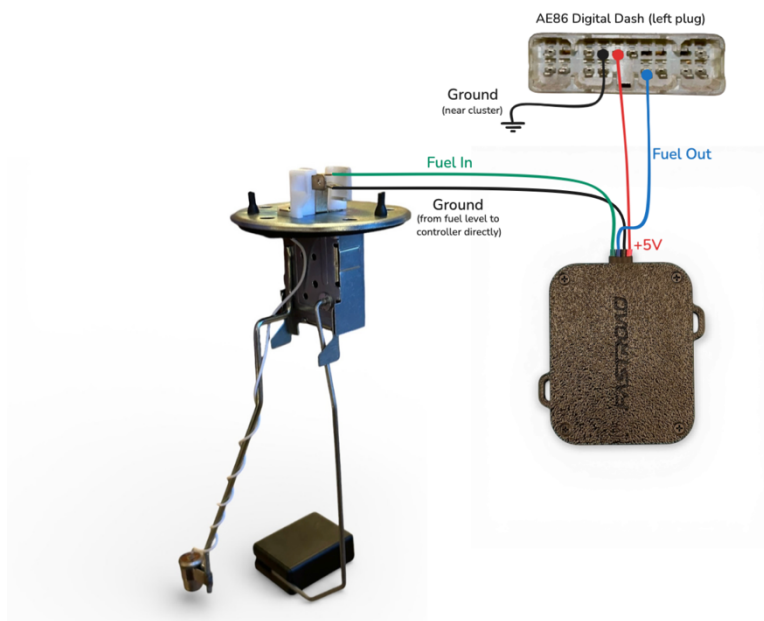


Figure 1 - Example of wiring in a Toyota AE86 Digital Cluster

(Optional for custom values)

1. Locate the stock fuel sender connector (usually near the fuel tank or under the rear seat on AE86/KE70).
2. Disconnect the original 3-pin sender plug.
3. Remove the fuel level sender from the tank to freely adjust the float between 0%, 50%, and 100% positions and use the app to calibrate the values.

5. First Power-Up and App Access (After Wiring)

Once you have connected all wires correctly (**5V**, **GND**, **FUEL IN**, **FUEL OUT**) and powered everything on, the module will start up and create its own WiFi access point for configuration (a blue led in the module will indicate the wifi is on)

Accessing the Web Application for calibration:

1. On your smartphone (or any WiFi-capable device), look for a new WiFi network named: **Fuel-Gauge-Controller**
2. Connect to it using the password: **gOr2tyMH1nHM**

(Tip: Scan the QR code below with your phone's camera to connect)



Note: Your phone will lose internet access while connected to this access point — that's normal. It doesn't need the internet to configure the module.

3. Once connected to the **Fuel-Gauge-Controller** network, open a web browser on your phone or computer and go to this address: <http://192.168.4.1/>
4. (Again, you can scan the QR code below to open the page instantly.)

Note: You may need to toggle Wi-Fi off and back on (or disable mobile data) on your current device. Some phones automatically disconnect after a while since this network has no internet access.



You should now see the module's configuration web app load. It looks like a simple dashboard with tabs for viewing calibration, wifi, etc.

5. **Default preset** By default, the module comes pre-programmed with the **AE86/KE70 preset**, this means it is already set up to read the stock Toyota sender (~3 Ω full $\rightarrow$ ~110 Ω empty) and output a signal compatible with most digital dashes expecting that range. If you're using it in a classic AE86/KE70 digital dash setup, you can often skip custom programming and go straight to testing.

6. Calibration: Custom Preset

Before you start, make sure to do all initial configuration and calibration **with the car stationary** and in a well-ventilated area. Also, make sure the lights, or any load, is turned off during the calibration since it can affect the readings.

1. In the app go to **Calibration** tab
2. Tap the **+** icon to create a new preset, give it a name and start calibration
3. **0% (Empty)**
 - Move float arm to empty position
 - Set the shown resistance for 0% in the app and save point
4. **50% (Half)**
 - Move float arm to halfway
 - Set the shown resistance for 50% and save point
5. **100% (Full)**
 - Move float arm to full position
 - Set the shown resistance for 100% and save point
6. **Save** the preset → go back to main screen
7. **Test**
 - Move the float arm and check if % changes in the app
 - Turn cluster off and on (in the case of AE86 digital cluster), the gauge should now show correct level (E → ½ → F)
 - Do a final test with the fuel level meter in the fuel tank, it should show the correct level of fuel.

IMPORTANT NOTE: In the AE86 digital cluster can take up to 5 minutes to update the fuel level after a change.

Quick fix: Turn the cluster **off and on** (ignition or power cycle) to instantly show the updated value. (If the app disconnects during the power cycle, it will reconnect automatically in a few seconds.)

7. Disable WiFi (Optional)

Turning off WiFi prevents unauthorized access to the controller and protects your presets. **This is intended as a permanent change for when you no longer need remote access.**

To disable WiFi go to the WiFi tab in the application and click the red button “Disable WiFi”.

IMPORTANT NOTE: This action is Irreversible via software: Once disabled, the web interface is no longer accessible wirelessly. WiFi can only be re-enabled by holding the BOOT button on the board for 6 seconds.

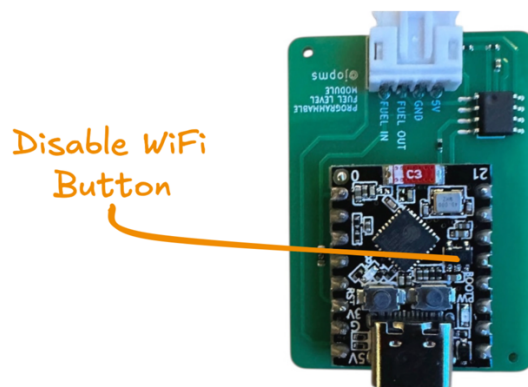


Figure 2 - Controller with BOOT (WiFi reset) button highlighted

8. Troubleshooting & Support

- Wifi not showing → Check blue led in the board, if not present check point 7 to enable WiFi again
- Dashboard says “Connecting” → Check if device is connected to the correct network
- Fuel reads backwards → Swap calibration points or reprogram output range
- No reading → Check power (5V present?), ground, and connections
- Erratic reading → Ensure good ground is being supplied; check sender float arm moves freely, and is in good condition.
- Dash shows full/empty only → Recalibrate or verify output range matches dash expectation.

For any other problems please reach out at info@fastroad-shop.com.

9. Firmware Updates

Updates are done through the app.

Easy way:

1. Go to **WiFi** section in the app.
2. Connect the module to your home WiFi.
3. Go to **Settings**.
4. Check for updates and install if available.

Manual way:

- Download the file from: <https://raw.githubusercontent.com/jopms/fuel-gauge-releases/main/firmware.bin>
- In the app → Settings → Upload the file → install.

After update, the module restarts. Check calibration if needed.

Enjoy, and Safe driving!

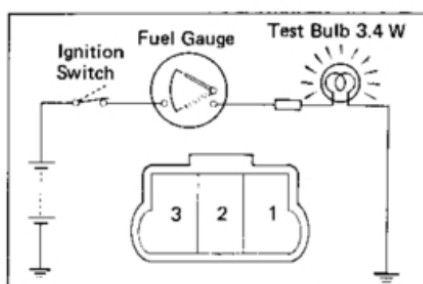
10. Appendix

Fuel Gauge

INSPECTION OF FUEL GAUGE

1. INSPECT RECEIVER GAUGE OPERATION

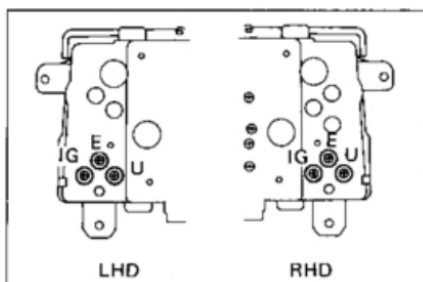
- Disconnect the connector from the fuel sender gauge. Turn the ignition switch on and check that the receiver gauge needle moves to empty position.



- Connect the 3.4W test bulb between terminal 1 and body ground. Check that the bulb light and the receiver gauge needle operates.

NOTE: Because of the silicon oil in the gauge, it will take about 90 seconds for the needle to stabilize.

If indications are not correct, remove and test the receiver gauge.



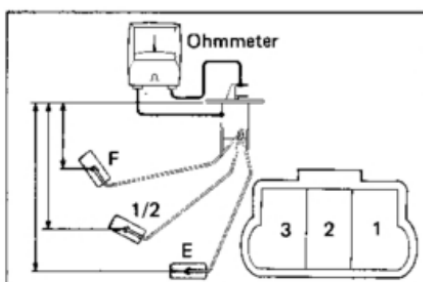
2. MEASURE RECEIVER GAUGE RESISTANCE BETWEEN TERMINALS

Between terminals	Resistance (Ω)
IG — U	Approx. 102
U — E	Approx. 101
IG — E	Approx. 203

If each resistance value is not as shown in the table above, replace the receiver gauge.

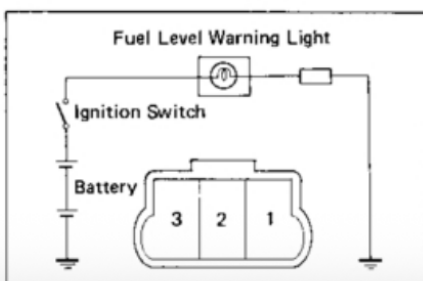
3. MEASURE RESISTANCE OF SENDER GAUGE

- Check that resistance changes as the float is moved from the top to bottom position.
- Measure the resistance between terminal 1 and the sender gauge body for each float position.



	Float position mm (in.)	Resistance (Ω)
F	64.7 — 70.7 (2.547 — 2.783)	3 ± 2.1
1/2	153 (6.024)	32.5 ± 4.8
E	205.5 — 211.5 (8.091 — 8.327)	110 ± 7.7

If each resistance value is not as shown in the table above, replace the sender gauge.



Fuel Level Warning

INSPECTION OF FUEL LEVEL WARNING

1. INSPECT WARNING LIGHT OPERATION

- Disconnect the connector from the switch. Connect the switch terminal 1 and body ground.
- Turn the ignition switch on. Check that the bulb lights.

If operation is not correct, remove and test the bulb.

Figure 3 - Picture from Toyota AE86 Manual

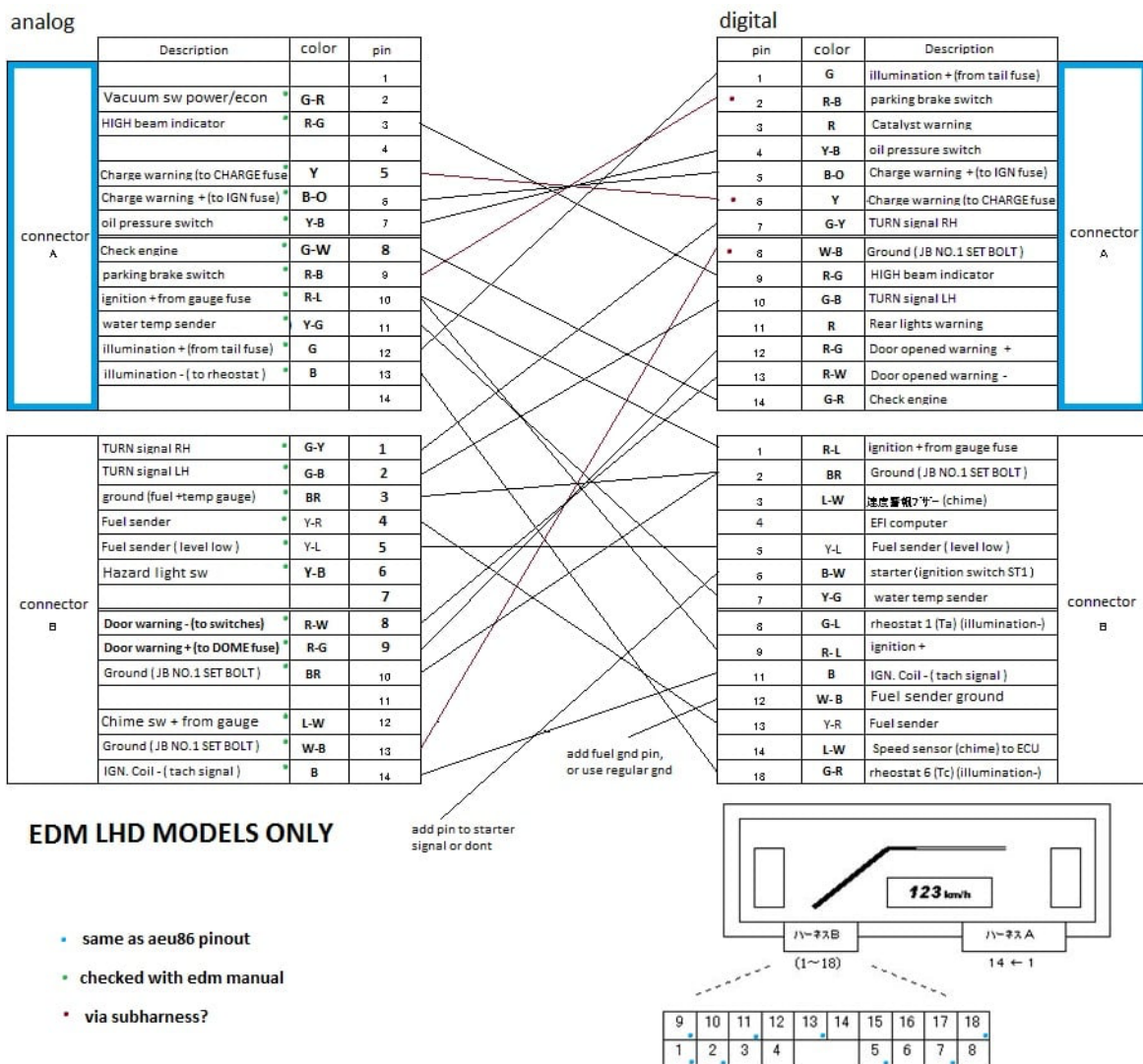


Figure 4 - Schematic to convert analog to digital cluster for Toyota AE86

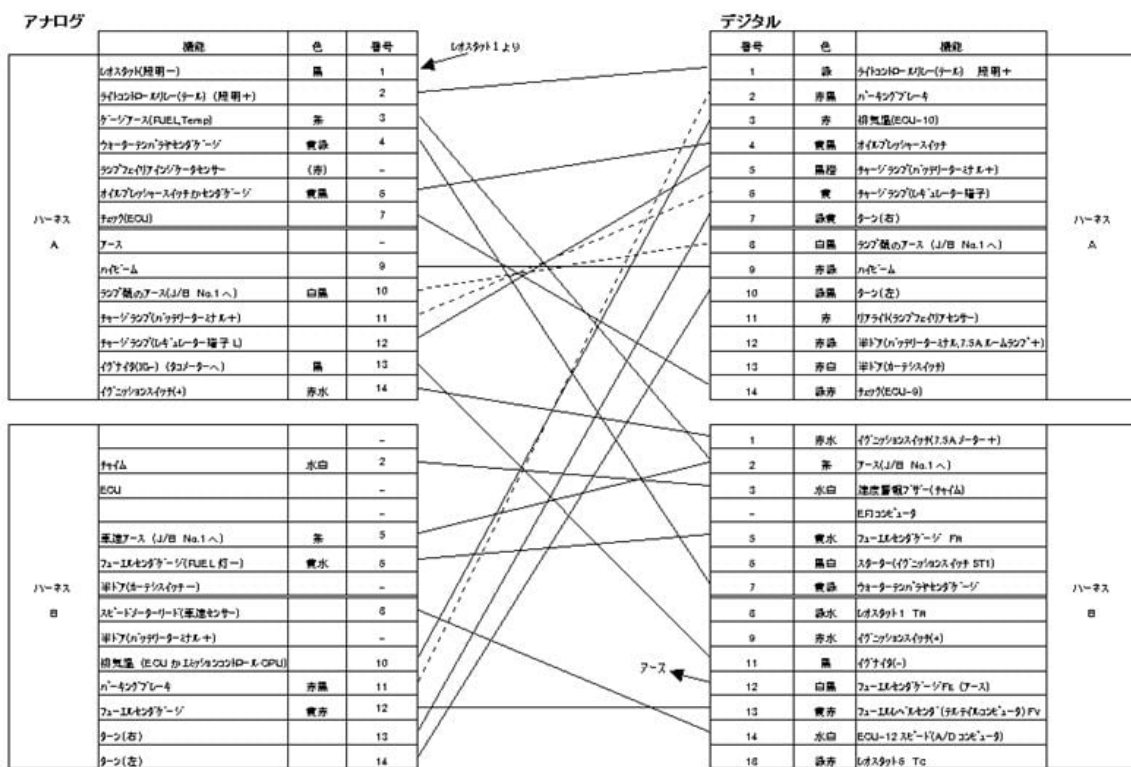


Figure 5 - Schematic to convert analog to digital cluster for Toyota AE86 (Japanese version)